

MATH 342 Notes

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1 First Order Vector-Valued ODE

1.1 Repeated eigenvalues

1. Example: Find a fundamental set for $\dot{x} = \begin{bmatrix} 1 & -1 \\ 4 & 5 \end{bmatrix} x$.

$$\lambda = 3, \quad \xi = [1, -2]^T \quad (1)$$

$$(d/dt - \lambda)^2 x = 0 \text{ has solns } Ce^{\lambda t}, Cte^{\lambda t} \quad (2)$$

$$(3)$$

Propose

$$x = te^{\lambda t} \xi ? \quad (4)$$

but then

$$\dot{x} - Ax = e^{\lambda t} \xi \neq 0 \quad (5)$$

so x is not a solution.

However, we can cancel off the nonhomogeneous part with an ansatz solution $e^{\lambda t} \eta$, $\eta \in \mathbb{R}^2$, to the nonhomogeneous equation $\dot{x} - Ax = e^{\lambda t} \xi$. (Compare with method of undetermined coefficients.)

So

$$x = te^{\lambda t} \xi - e^{\lambda t} \eta. \quad (6)$$

Then $\dot{x} - Ax = 0$ gives

$$0 = e^{\lambda t} \xi + (\lambda e^{\lambda t} \eta - Ae^{\lambda t} \eta), \quad (7)$$

and therefore

$$e^{\lambda t} (\xi + \lambda \eta - A\eta) = 0 \implies \xi + \lambda \eta - A\eta = 0 \quad (8)$$

$$\implies \xi = (A - \lambda I)\eta. \quad (9)$$

Notice that $A - \lambda I$ is not invertible. But concretely we want to solve

$$\begin{bmatrix} -2 & -1 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} \eta_1 \\ \eta_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}. \quad (10)$$

Is ξ always in the column space of $A - \lambda I$?

2. Summary (choice of a fundamental set).

In general when A has repeated eigenvalues, and A is 2×2 ,

$$x^1 = e^{\lambda t} \xi, \quad x^2 = te^{\lambda t} \xi + e^{\lambda t} \eta, \quad (11)$$

where η is a solution to $(A - \lambda I)\eta = \xi$.

η is called a *generalised eigenvector*.

Highly important result. If η is *any* vector independent of ξ , we will have $(A - \lambda I)\eta = c\xi$ for some $c \neq 0$. (Why?)

1.2 Nonhomogeneous system

1. The system of interest is

$$\dot{x} = P(t)x + g(t), \quad P(t) \in M_{n \times n}(\mathbb{R}), \quad x(t), g(t) \in M_{n \times 1}(\mathbb{R}) \quad (12)$$

Solution is given as a sum of a particular solution with a general solution of the corresponding homogeneous system (which we have already done). We will focus on finding particular solutions x_p to (12).

2. Methods:

- (a) Undetermined coefficients for special $g(t)$
- (b) Laplace transforms
- (c) Decoupled systems for diagonalizable A
- (d) **Variation of parameters** (we will use this one)

Example: $\dot{x} = \begin{bmatrix} 1 & -1 \\ 4 & 5 \end{bmatrix} x + \begin{bmatrix} e^t \\ 0 \end{bmatrix}.$

3. Proposed solutions (VOP):

$$x(t) = u_1(t)x^1(t) + u_2(t)x^2(t) \quad (\text{undetermined coefficients}) \quad (13)$$

where u_1, u_2 are scalar valued and x^1, x^2 solve the homogeneous equation. The goal is to find u_1, u_2 .

- (a) Rewrite

$$x(t) = \begin{bmatrix} x^1(t) & x^2(t) \end{bmatrix} \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix} \quad (14)$$

$$= \Psi(t)u(t) \quad (15)$$

$$= \begin{bmatrix} e^t & e^{5t} \\ -3e^t & e^{5t} \end{bmatrix} \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix} \quad (16)$$

- (b) Apply $x(t) = \Psi(t)u(t)$ to $\dot{x}(t) = Ax(t) + g(t)$ to get

$$\Psi'(t)u(t) + \Psi(t)u'(t) = A\Psi(t)u(t) + g(t) \quad (17)$$

- (c) Remember $\Psi(t)$ is invertible, and also explicitly

$$\Psi(t) = \begin{bmatrix} x^1(t) & x^2(t) \end{bmatrix} \quad (18)$$

$$\Psi'(t) = \begin{bmatrix} Ax^1(t) & Ax^2(t) \end{bmatrix} \quad (19)$$

$$= A\Psi(t). \quad (20)$$

Ψ solves a “matrix version” of the original homogeneous equation. We use this result to simplify (17):

$$\Psi(t)u'(t) = g(t). \quad (21)$$

(d) We have

$$\begin{bmatrix} e^t & e^{5t} \\ -3e^t & e^{5t} \end{bmatrix} \begin{bmatrix} u'_1(t) & u'_2(t) \end{bmatrix} = \begin{bmatrix} e^t & 0 \end{bmatrix}, \quad (22)$$

so

$$\begin{cases} e^t u'_1(t) + e^{5t} u'_2(t) = e^t \\ -3e^t u'_1(t) + e^{5t} u'_2(t) = 0 \end{cases} \quad (23)$$

A solution is

$$\begin{cases} u'_1(t) = 1/4 \\ u'_2(t) = \frac{3}{4}e^{-4t} \end{cases} \quad (24)$$

(e) Recover $u(t)$ from $u'(t)$.

$$u(t) = \int u'(t)dt = \int \begin{bmatrix} \frac{1}{4} \\ \frac{3}{4}e^{-4t} \end{bmatrix} dt = \begin{bmatrix} \frac{1}{4}t \\ -\frac{3}{16}e^{-4t} \end{bmatrix} + c \quad (25)$$

(f) One particular solution is

$$x(t) = \left(\int_0^t \frac{1}{4} ds \right) \begin{bmatrix} e^t \\ -3e^t \end{bmatrix} + \left(\int_0^t \frac{3}{4} e^{-4s} ds \right) \begin{bmatrix} e^{5t} \\ e^{5t} \end{bmatrix}. \quad (26)$$

4. *Solution method in general.*

Problem

$$\dot{x}(t) = P(t)x(t) + g(t).$$

Corresponding homogeneous system

$$\dot{x}(t) = P(t)x(t),$$

solved by

$$\dot{x}_h = P(t)x_h.$$

Solution to the nonhomogeneous system

$$x(t) = \Psi(t)u(t)$$

(solve for $u(t)$).

Equation for $u(t)$

$$\Psi'(t)u(t) + \Psi(t)u'(t) = P(t)\Psi(t)u(t) + g(t).$$

OBSERVE that $\Psi'(t) = P(t)\Psi(t)$ (even in the nonconstant $P(t)$ case), so that

$$\Psi(t)u(t) = g(t).$$

Furthermore, Ψ is invertible. This is because of the whole Wronskian nonzero $\implies$ nonzero everywhere / diffeq's do not collapse the dimension of subspaces spanned by solutions thing. Thus

$$u(t) = \int \Psi(t)^{-1} g(t) dt \quad (27)$$

$$\implies x_p(t) = \Psi(t) \left(c + \int_0^t \Psi(s)^{-1} g(s) ds \right) \quad (28)$$

$$= \Psi(t)c + \int_0^t \Psi(t)\Psi(s)^{-1} g(s) ds, \quad (29)$$

(which is even the general solution to the original equation, with degree of freedom c to be determined by initial conditions).

Special case: $P(t) = A$ (constant). Then

$$x(t) = e^{tA}x(0) = \int_0^t e^{(t-s)A} g(s) ds \quad (\text{Duhamel's formula}) \quad (30)$$

2 Power Series Solutions to ODE

2.1 Basic power series tools

1. Example.

$$f(x) = \sum_{n=1}^{\infty} a_n x^n = \sum_{n=1}^{\infty} \frac{2^n}{n} x^n. \quad (31)$$

(a) Find the radius of convergence.

The radius of convergence of a power series $f(x) = \sum_{n=1}^{\infty} a_n (x - x_0)^n$ is a number $R \geq 0$ such that

- i. $\forall x$ satisfying $|x - x_0| < R$, $f(x)$ converges absolutely (as in, $\sum_{n=1}^{\infty} |a_n| |x - x_0|^n$ converges).
- ii. $\sum_{n=1}^{\infty} a_n (x - x_0)^n$ diverges whenever $|x - x_0| > R$.

R is called the *radius of convergence*.

(Here $R = 1/2$ by the Ratio Test.)

Tools for finding the radius of convergence

The Root Test: If $\lim_{n \rightarrow \infty} |a_n|^{1/n}$ converges, then $R = \frac{1}{\lim_{n \rightarrow \infty} |a_n|^{1/n}}$.

For instance, (31) gives

$$R = \frac{1}{\lim_{n \rightarrow \infty} (2^n/n)^{1/n}} = \frac{1}{\lim_{n \rightarrow \infty} 2/n^{1/n}} = 1/2 \quad \text{since } n^{1/n} \rightarrow 1 \quad (32)$$

The Ratio Test: If $\lim_{n \rightarrow \infty} \frac{|a_{n+1}|}{|a_n|}$ converges, then $R = \frac{1}{\lim_{n \rightarrow \infty} \frac{|a_{n+1}|}{|a_n|}}$.

In our example,

$$R = \frac{1}{\lim_{n \rightarrow \infty} \frac{2^{n+1}/(n+1)}{2^n/n}} = \frac{1}{\lim_{n \rightarrow \infty} 2n/n+1} = 1/2. \quad (33)$$

(b) Find $f'(x)$ (for meaningful x)

As usual, differentiate term by term. If $f(x)$ is given by a power series $\sum_{n=0}^{\infty} a_n x^n$, then

$$f'(x) = \sum_{n=0}^{\infty} n a_n x^{n-1} = \sum_{n=1}^{\infty} n a_n x^{n-1} \quad (34)$$

By making a change of variable $k = n - 1$, and noticing that the first term drops out of the above series (because $n = 0$), we have

$$f'(x) = \sum_{k=0}^{\infty} (k+1) a_{k+1} x^k. \quad (35)$$

This procedure for getting the power series into the desired powers of x is called *index shifting*. It is important to keep track of the initial value for k and n in the sum. This is analagous to changing the bounds of integration when performing a change of variable.

A remarkable feature of the radius of convergence is that it is *invariant under differentiation or integration of the power series*. Consider our example. Differentiating term by term or using (35), the derivative series is

$$f'(x) = \sum_{k=0}^{\infty} (k+1) \frac{2^{k+1}}{k+1} x^k = \sum_{n=0}^{\infty} 2^{n+1} x^n. \quad (36)$$

By the ratio test, the radius of convergence of $f'(x)$ is

$$R = \frac{1}{\lim_{n \rightarrow \infty} \frac{2^{n+2}}{2^{n+1}}} = \frac{1}{2}, \quad (37)$$

which is the same as the radius of convergence for $f(x)$. The radius of convergence of the integral power series $\int dx f(x)$ is a fun exercise (it should also be $1/2$).

(c) Some notions. Let

$$f(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n \quad \text{with} \quad R > 0. \quad (38)$$

- i. $f(x)$ is *analytic* at $x = x_0$. (This is what it means to be analytic at x_0 ; there exists a convergent power series equal to the function on a neighborhood of x_0 .)
- ii. $a_n = f^{(n)}(x_0)/n!$, and $f(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + \dots$, and in particular $f(x_0) = a_0$.
- iii. Taylor series expansion: When f is analytic at x_0 , it has a particular power series representation for $f(x)$, given by

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n \quad (39)$$

For example, the exponential function is its own derivative and $e^0 = 1$. This is enough to define the exponential everywhere by its Taylor series at $x_0 = 0$:

$$e^x = \sum_{n=0}^{\infty} \frac{(e^x)^{(n)}|_{x=0}}{n!} x^n = \sum_{n=0}^{\infty} \frac{e^0}{n!} x^n = \sum_{n=0}^{\infty} \frac{1}{n!} x^n, \quad (40)$$

which should look familiar.

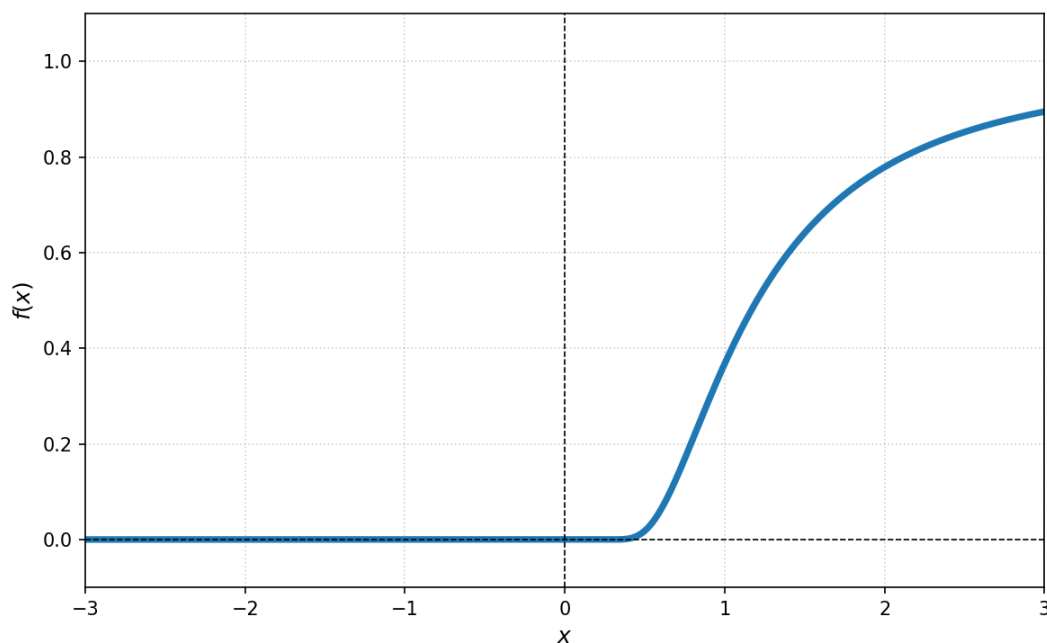
2. Analytic vs infinitely differentiable functions.

Question: Every function that is analytic at $x = 0$ is infinitely differentiable at $x = 0$. Is the converse true? NO!

Example: A classic example is the function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$f(x) = \begin{cases} e^{-1/x^2}, & x > 0 \\ 0, & x \leq 0 \end{cases} \quad (41)$$

This function is infinitely differentiable at $x_0 = 0$ (why?) but is not analytic there. For if it were analytic, its Taylor series would be identically zero since $f^{(n)}(0) = 0$ for all n (look at the value of f for negative x arbitrarily close to zero).



Fact: $\sum_{n=0}^{\infty} a_n x^n = 0 \implies a_n = 0$ for all n . This is just a restatement of what we said above. (If you define $f(x)$ by that power series then we identify the a_n with the derivatives of f , which are all zero since f is by assumption identically zero.)

2.2 Homogeneous DE with variable coefficients

Consider the general type of ordinary differential equation

$$P(x)y'' + Q(x)y' + R(x)y = 0 \quad (y = y(x) \text{ is real valued}) \quad (42)$$

We will make the following assumption.

Assumption: P, Q, R are polynomials and there is no scalar c such that $P(x), Q(x), R(x)$ are all divisible by $(x - c)$ (they are essentially “coprime” as elements of the ring of polynomials over the real or complex numbers).

1. Well known equations.

(a) Airy’s equation:

$$y'' - xy = 0, \quad P(x) = 1, Q = 0, R(x) = -x. \quad (43)$$

(b) Bessel's equation:

$$x^2 y'' + xy' + (x^2 - \nu^2)y = 0, \quad P(x) = x^2, Q(x) = x, R(x) = x^2 - \nu^2 \quad (44)$$

where ν (the Greek letter “nu”) is a parameter in the equation.

If $\nu \neq 0$, then our assumption (about the coprime nature of P, Q, R) is satisfied, but not if $\nu = 0$. In the case $\nu = 0$, the equation simplifies a fair bit.

(c) Legendre's equation:

$$(1 - x^2)y'' - 2xy' + \alpha(\alpha + 1)y = 0, \quad (45)$$

where α is a parameter.

2. Solution methods.

Let us restate our DE in **standard form**.

$$y'' + \frac{Q}{P}(x)y' + \frac{R}{P}(x)y = 0 \quad (46)$$

or

$$y'' + p(x)y' + q(x)y = 0, \quad (47)$$

where $p(x) = \frac{Q}{P}(x)$, and $q(x) = \frac{R}{P}(x)$.

But wait! We need to verify that it is okay to divide by $P(x)$ (what if we are dividing by zero?). If $P(x) \neq 0$, then $P(x) \neq 0$ for all x near x_0 , and the standard form is valid near x_0 . In this situation, x_0 is called an **ordinary point**.